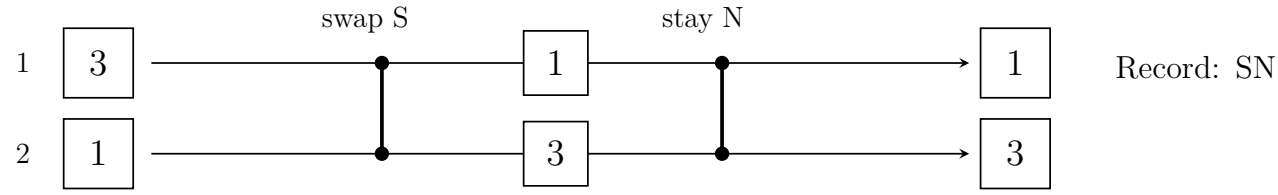
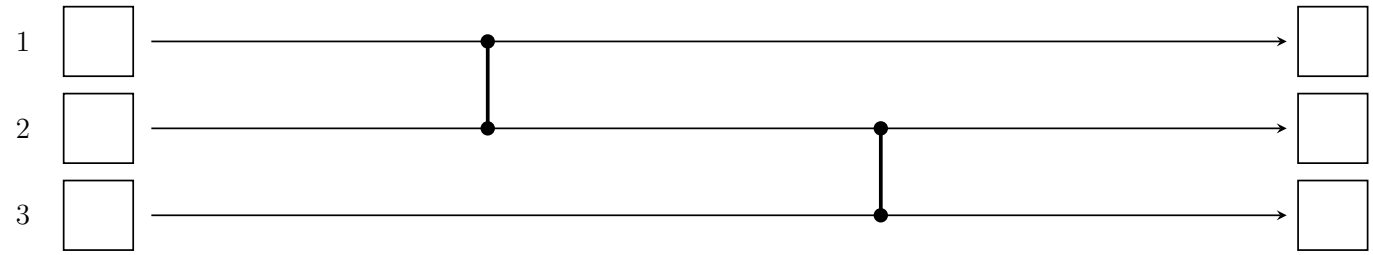
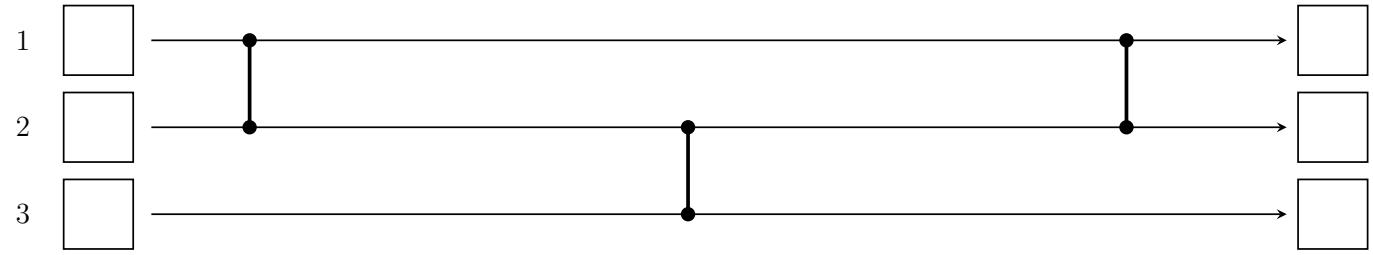
Move from left to right. At each bar, compare only the two lanes with black dots: the smaller value goes to the upper lane and the larger to the lower lane. Ties may stay put. Keep every bar fixed for every start; move cards only at bars. A machine sorts when the final values never get smaller from lane 1 down.

Example: compare the same two lanes twice.



Problem 6: Write S when two cards swap at a bar and N when they stay put, in order from left to right. For each machine, find every record made by orders of 1, 2, 3, and find every start that makes each record.





Could a fixed machine sort two different starting orders of distinct cards while making the same swap/no-swap record? Explain.

Problem 7: Find the fewest bars a three-lane sorting machine can have and the fewest bars a four-lane sorting machine can have. Give a working machine for each and explain why fewer bars cannot work.

